



Department	Fast School of Management	Dept. Code	BBA, BSAF, BSBA
Course Title	Business Mathematics 2	Course Code	MT 2004
Pre-requisite(s)	Business Mathematics 1	Credit Hrs.	3
Moderator	Dr. Hina Firdous		
Course Instructors	Dr. Hina Firdous,		
Note:	It is a tentative schedule of course. It may vary (if required).		

Course Objective	Business Mathematics presents Mathematical skills and knowledge that you can apply to solve financial problems. Finance plays a major role in our lives e.g. car loans, mortgage payments, retirement plans, real estate investment and knowing how to calculate the cost of borrowing or the return on investment is important to us. This course provides step-by-step guidance through sample problems and solutions related to banking, credit, basic finance and investment. Through this course students can gain an understanding of financial instruments and terminology used in business finance such as compound interest, annuities and promissory notes. This course also gives knowledge about how to use calculus in the field of business and management. The course serves as a good foundation for further study in management, accounting and finance.
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I = Introduction, R = Reinforcement, E = Evaluation.

A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Tools
01	<i>To understand the basic Non Linear function and its applications in Business and Economics</i>	A1,Q1
02	<i>To learn Linear Programming and its importance</i>	A1,Q1,M1
03	<i>To understand Financial Mathematics Annuities, Debt Repayments, Sinking funds</i>	A2, Q2



04	<i>Learn Optimization of Bivariate functions and its applications for Business students.</i>	A2,Q2,M2
05	<i>Apply Integral calculus to problems in Economics</i>	A3, Q3,F

Text Book(s)	Title	<i>Applied Mathematics for Business, Economics, and Social Sciences</i> , 4 th Ed., Singapore
	Author	Frank S. Budnick
Ref. Book(s)	Title	<i>Essential Mathematics for Economics and Business</i> , 3 rd Ed.
	Author	Teresa Bradley

Week	Course Contents	Chapter	CLO
01	Brief Synopsis of the Course, Quadratic function, Exponential function, Logarithmic function Applications: supply, demand, revenue, cost	6,7	1
02	Alternative Optimal Solutions, No Feasible Solution, Unbounded Solution. The Simplex Method (Maximization and Minimization Problems)	11	2
03	Application of Linear Programming, Phenomena of alternative optimal solutions, no feasible solution, Unbounded solution.	11	2
04	Depreciation, Annuities and their future value	8	3
05	MID-TERM-I		
06	Annuities and their present value	8	3
07	Partial Derivatives, Optimization of Bivariate Functions	20	4
08	Application of Bivariate Optimization	20	4
09	Integration of transcendental functions Methods of integration	18	5
10	The definite integral and area under the curve	19	5
11	MID-TERM-II		
12	Methods of Approximation Rectangle Rule,	19	5
13	Trapezoidal Rule, Simpson's Rule		



14	Application of Integral Calculus	19	5
15	Differential Equations	18	5
	FINAL EXAM		

Evaluation Procedure & Marks Distribution:

Assessment Tools	Total No. of	Weightage
Quizzes	Minimum 3	10%
Assignments	Minimum 3	10%
Mid Term Exam	2 (I+II)	30%
Final Exam	1	50%

Note: No homework/ assignment submission after due date. Relative grading policy will be followed.